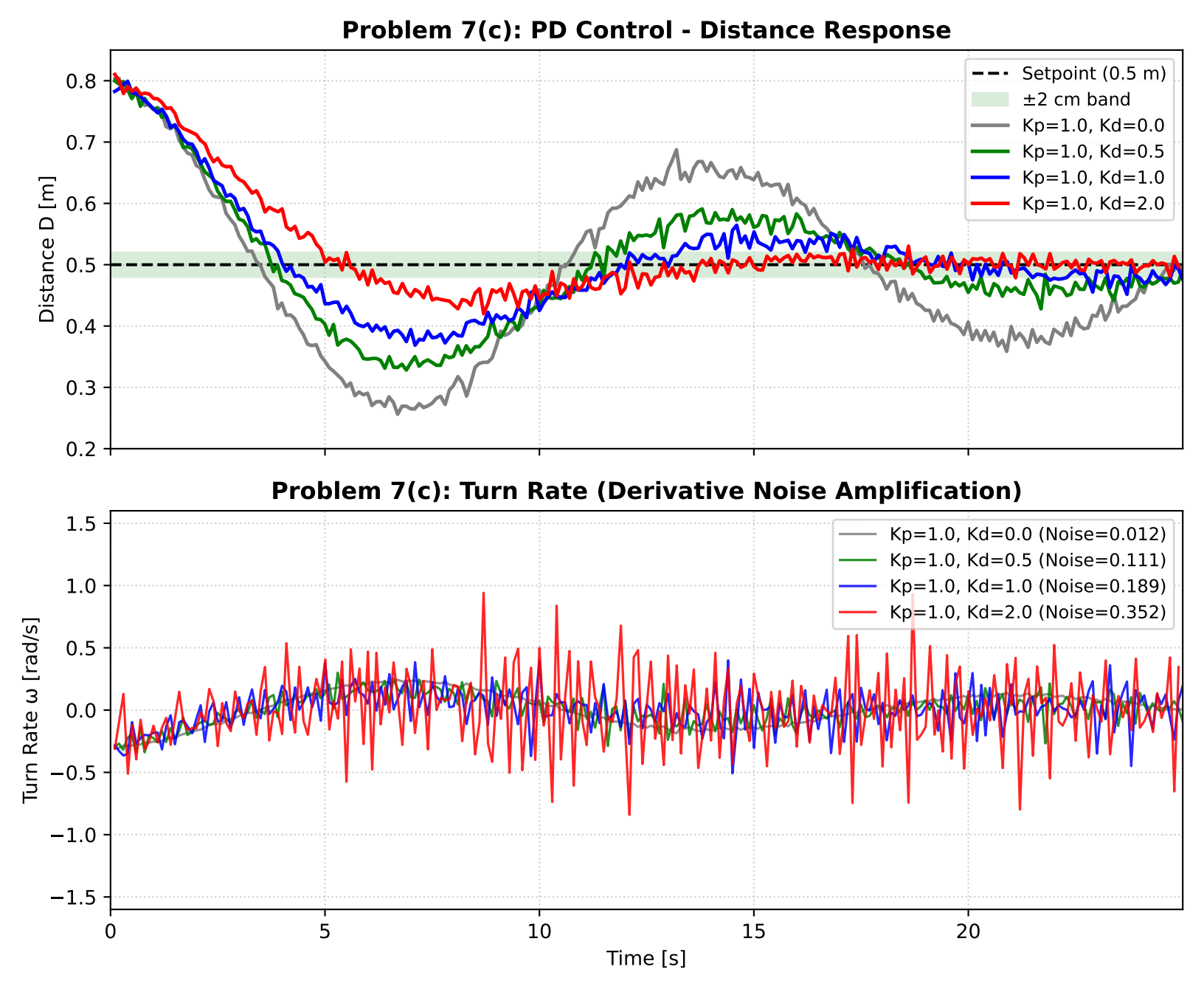
Problem 7(c): PD Control - Distance Response
Distance D [m]
0.8
0.7
0.6
0.5
0.4
0.3
0.2
Setpoint (0.5 m)
±2 cm band
Kp=1.0, Kd=0.0
Kp=1.0, Kd=0.5
Kp=1.0, Kd=1.0
Kp=1.0, Kd=2.0
Problem 7(c): Turn Rate (Derivative Noise Amplification)
Turn Rate ω [rad/s]
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Kp=1.0, Kd=0.0 (Noise=0.012)
Kp=1.0, Kd=0.5 (Noise=0.111)
Kp=1.0, Kd=1.0 (Noise=0.189)
Kp=1.0, Kd=2.0 (Noise=0.352)
0
5
10
15
20
Time [s]